HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
Faculty for High Quality Training

Program name: Automation and Control Engineering Technology

FINAL EXAMINATION
SEMESTER 2 – ACADEMIC YEAR 2021-2022
Course name: Applied Mathematical in Electrical Engineering

Course ID: AMEE331944E
Exam code: 01 Number of pages: 02
Duration: 90 minutes.
Used paper documents.

Question 1 (2.5 marks): Given the linear circuit in Figure 1, show the details of your work in

a) modelling the linear circuit by three equations in three unknown branch currents i_1, i_2, i_3 by using Kirchhoff's law,

b) calculating the determinant of the coefficient matrix $\boldsymbol{A}$, finding the inverse matrix $\boldsymbol{A}^{-1}$ by determinants and obtaining i_1, i_2, i_3 by using inverse matrix $\boldsymbol{A}^{-1}$.

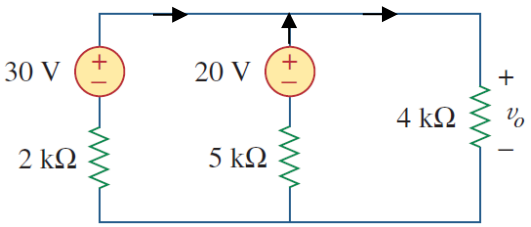

Figure 1

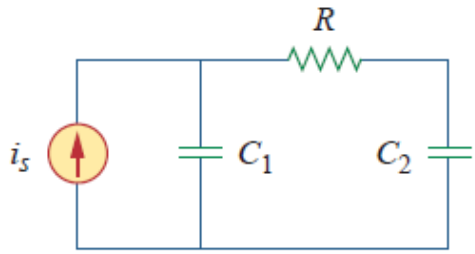

Figure 2

Question 2 (2.5 marks): Given the circuit in Figure 2 with $R = 20\ \Omega, C_1 = 0.01\ F, C_2 = 0.05\ F$, $i_s(t) = 5\ cost\ 5t$ A if $0 \le t < \infty$. Show the details of your work in

a) modelling this circuit by second order ordinary differential equation in unknown voltage v_2,

b) determining the steady-state voltage v_{2p},

c) finding a general solution of the second order ordinary differential equation v_2.

Question 3 (2.5 marks): Given the circuit in Figure 3, $R_1 = 10\Omega, R_2 = 2\Omega, L_1 = 1\ H, L_2 = 2\ H$, $v_s = 100$ V if $0 \le t < \infty$. Show the details of your work in

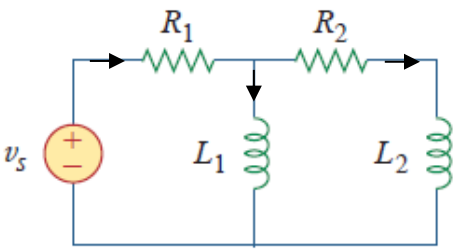

Figure 3

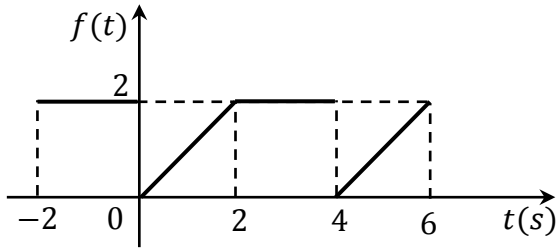

Figure 4

a) modelling this circuit by two linear ordinary differential equations in two unknown voltage i_1 and i_2,

b) determining the two equivalent equations by using the Laplace transform with $i_1(0) = 0$, $i_2(0) = 2\ A$,

c) finding I_1 by using the Cramer's rule and $i_1(t)$ by using the inverse Laplace transform.

Question 4 (2.5 marks): Given the periodic function $f(t)$ in Figure 4, show the details of your work in

a) finding the Fourier coefficients a_0, a_n and b_n of the function $f(t)$,

b) calculating the amplitude A_n and phase φ_n if $Z_n = A_n\angle\varphi_n = a_n - jb_n$

c) graphing Z_1, Z_2 and Z_3 in the complex plane.

Note: Proctors are not allowed to give any unauthorized explanations.

Learning outcome mapping	**Assessed in**
[LO 1.1]: Modelling of electrical engineering problems by a linear system of equations and solving by using matrix theories.	Questions 1
[LO 2.1]: Modelling of electrical engineering problems by a second order ordinary differential equation and solve.	Questions 2
[LO 2.4]: Modelling of electrical engineering problems by ordinary differential equation systems and solving by using Laplace transform.	Question 3
[LO 2.5]: Performing analyses with complex numbers	Question 4
[LO 2.3]: Performing analyses with Laplace transform	Question 3
[LO 2.6]: Analysis and argumentation to solve problems by Fourier series	Question 4

1/06/2022

Approved by Program Chair

(signed and named)